Dynatrace Incident Response SOP

SOP-001: Availability Alert Remediation (Host Down / Process Crash)

Event Type: Availability
Severity Level: SEV-1 (Critical)
Impacted Entities: Linux/Windows Hosts, Core Microservices

Diagnostic Steps

1. Open the Dynatrace Problem Feed and isolate the active Problem ID. Verify the Root Cause block.

2. Navigate to the Host Infrastructure Overview to verify hypervisor/hardware status.

3. Check for concurrent Resource Alerts (e.g., CPU, Memory Exhaustion, Disk Full).

4. View specific process lifecycle logs to see if an OOM Kill (Exit Code 137) or sudden crash occurred.

Remediation Steps

- If Process OOM: Execute an automated service restart via Kubernetes/Ansible. Adjust memory requests/limits if structural.

- If Host Unreachable: Trigger an instance reboot via cloud console (AWS EC2 / Azure VM) or escalate to infrastructure team.

- Verification: Confirm entity status switches back to green and the Davis Problem automatically closes.

SOP-002: Service Performance Degradation & Slowdown Triage

Event Type: Slowdown
Severity Level: SEV-2 to SEV-3 (Major/Minor)
Impacted Entities: Web APIs, Microservices, Databases

Diagnostic Steps

1. Access the Response Time Degradation chart within the active Dynatrace Problem card.

2. Launch the Service Flow interface to trace request paths and locate downstream bottlenecks.

3. Drill down into Method-Level Hotspots to isolate long execution lines, database locks, or code loops.

4. Review SQL statements to check for high execution times or connection pool starvation issues.

Remediation Steps

- Downstream Bottleneck: Contact the respective service or DBA team; apply circuit breakers or caching if available.

- Sudden Load Spike: Horizonally scale application replicas/pods to manage the transaction surge.

- Code Release Regressions: If the slowdown aligns with a recent deployment marker, initiate an immediate rollback.

SOP-003: High Error Rate and Exception Spike Management

Event Type: Error
Severity Level: SEV-2 (Major)
Impacted Entities: Web Frontend Applications, API Gateway, Transaction Services

Diagnostic Steps

1. Review the Failure Rate Analysis dashboard inside the active problem card to isolate technical vs. functional errors.

2. Access the Exception Hub and review the stack trace details to discover failing lines of code.

3. Check correlated log events in Dynatrace Log Monitoring for warning/error lines near the timestamp.

4. Check User Session Analytics to see if errors are localized to particular browsers, geolocations, or OS versions.

Remediation Steps

- Code Defect Patching: Inform the development team with the exact stack trace for an emergency hotfix deployment.

- Authentication / Configuration: Correct misconfigured runtime variables, expired tokens, or credential entries.

- Faulty Nodes: Route traffic away or drain connections from corrupted hosts or instances to restore service.

SOP-004: Resource Contention & Capacity Saturation (CPU / Memory / Disk)

Event Type: Resource / Saturation

Severity Level: SEV-2 to SEV-3 (Major/Minor)
Impacted Entities: Hosts, Container Nodes, Database Servers

Diagnostic Steps

1. Access the Dynatrace Problem Card and open the Host Infrastructure / Container Overview panel to identify the specific metric exceeding thresholds (e.g., CPU % Usage, Memory Consumption, Disk Space %).
2. Analyze the Process Performance breakdown to pinpoint the specific application process, memory leak, or runaway thread causing resource exhaustion.
3. Check the Disk IOPS / Storage Analysis tab for high I/O wait times or log file bloat in mounted volumes.
4. Review historical baseline graphs over the last 7 to 30 days to determine if the spike is a predictable load cycle or an anomaly.

Remediation Steps

- Disk Cleanup / Volume Expansion: Truncate or archive non-essential logs, clear temporary files, or request an online cloud volume expansion.
- Resource Reallocation: Increase CPU/Memory allocations (vertical scaling) or scale deployment replicas (horizontal scaling) via Kubernetes/Terraform.
- Process Isolation: Kill or restart unresponsive/runaway worker processes consuming disproportionate resources.

SOP-005: Synthetic & RUM Infrastructure Monitoring Failures

Event Type: Availability / Performance (User Experience)
Severity Level: SEV-1 to SEV-2 (Critical/Major)
Impacted Entities: Web Frontends, Public APIs, Synthetic Monitors

Diagnostic Steps

1. Open the Synthetic Monitor Execution Details page for the failing location test to check screenshot capture and HTTP response status codes.
2. Review the Waterfalls Analysis to isolate DNS resolution time, TCP connect time, TLS handshake latency, or failing third-party scripts.
3. Check Real User Monitoring (RUM) Session Replays for impacted end-users to confirm if the issue is global or isolated to specific locations, ISPs, or device types.
4. Check external CDN/WAF status dashboards (e.g., Cloudflare, Akamai) to rule out edge-routing failures.

Remediation Steps

- CDN / Edge Routing Failover: Route traffic around degraded CDN locations or update edge routing rules.
- Third-Party Dependency Bypass: Disable or fallback non-critical third-party integrations (e.g., analytics tags, chat widgets) failing to respond.
- Certificate / Credentials Refresh: Update expired SSL/TLS certificates or authentication tokens configured within synthetic scripts.

SOP-006: Custom Event & Metric Alert Triggering

Event Type: Custom Event / Metric Threshold
Severity Level: SEV-3 to SEV-4 (Minor/Informational)
Impacted Entities: Custom Business Events, Log Analytics Alerts, Third-Party Telemetry

Diagnostic Steps

1. Open the Dynatrace Metric Explorer / Custom Charting page to verify the exact metric key and condition that breached the configured threshold.
2. Cross-reference the alert timestamp with Dynatrace Log Management using key extraction fields to trace custom application events.
3. Review the Anomaly Detection Rule Settings for the custom metric to confirm whether the alert was triggered by a static threshold or an adaptive baseline deviation.
4. Verify whether upstream changes or deployments introduced unexpected metric values or log formats.

Remediation Steps

- Threshold Tuning / Threshold Maintenance: Adjust sensitivity, sliding evaluation windows, or threshold boundaries if the alert is determined to be a false positive.
- Application Configuration Adjustment: Update feature flags, queue sizes, or rate limits tied to the custom business/application metric.
- Escalation to Domain Owners: Forward the alert context and custom metric payload to the specific product or engineering owner assigned to that functional domain.